Data stored

1) Accumulated adjacency

$\boldsymbol{A}_t \in \mathbb{R}^{n \times n}$

sparse floating-point tensor

$[\boldsymbol{A}_t]_{ij}$

edge weight for the indexed vertex pair



2) Optional vertex features

$\boldsymbol{X} \in \mathbb{R}^{n \times d}$

dense/sparse floating-point tensor

$[\boldsymbol{X}]_{ik}$ indexed feature value



3) Stored label hierarchy

$\boldsymbol{C}_t \in \mathbb{Z}^{L \times n}$

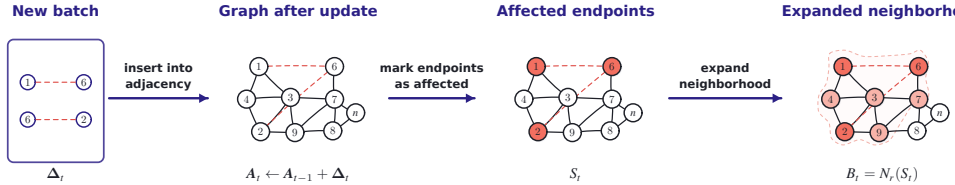
dense integer label tensor

$\ell \backslash v$	1	2	⋯	n
$L-1$	1	2	⋯	12
⋮	⋮	⋮	⋮	⋮
i	1	1	⋯	6
⋮	⋮	⋮	⋮	⋮
0	1	1	⋯	3

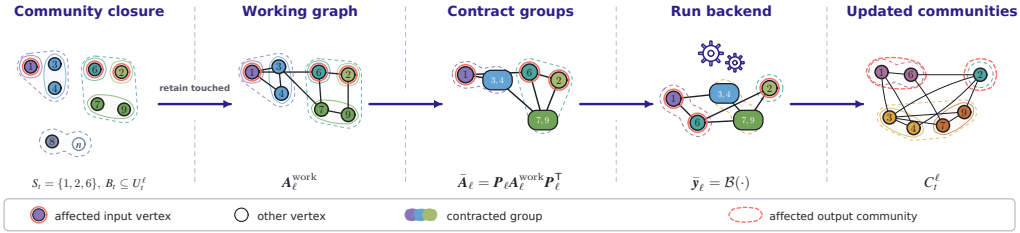
Backend configuration

feature-aware	topology-only
group sum or mean	adjacency only
$(\bar{\boldsymbol{A}}_\ell, \bar{\boldsymbol{X}}_\ell)$	$\bar{\boldsymbol{A}}_\ell$

1 Update and affected-region detection



2 Hierarchical local recomputation



3 Repeat backend calls across stored levels



Legend: edge — new / changed edge